K.B. Sendhil Kumar

Email: sendhilbalu@gmail.com
Phone: +91-8123867417, +91-8667860033

Experience Summary:

19.3 years of experience in Build to Specification Power Electronics Product Development in Aerospace and Defence, Automotive, Railways, Medical, Industrial and Commercial for Indian and Overseas Customers. Experience in handling the team and providing technical solutions, mentoring the fresher & juniors in the team. Involved in multiple end-end power electronics product development from concept to proto and proto to production.

Professional Experience Summary:

- Experience in **Power Electronics product Design and development from RFQ till manufacturing which includes specification Feasibility analysis, topology selection, component selection, schematic design, prototype design and verification & validation**
- In-depth knowledge on **SMPS, Fly back, Forward, Push-pull, Half-bridge, Full bridge converter** and **Power factor controller (PFC)** design
- In-depth knowledge on **Non-Isolated & Isolated Buck and Boost** Converter design
- Good knowledge in **Phase Shifted Full bridge** converter design
- **LLC Resonance based DC-DC converter design**
- **MOSFET, IGBT and eGaN based AC-DC, DC-DC Power Electronics Converter design for Inverter, UPS, Motor drives, battery charger etc**
- In-depth knowledge **on Sigle phase inverter and Three phase Inverter**
- **440V 250A High voltage and high current Solid State Circuit Breaker (SSCB) design and development, based on JFET for 3 phase AC and DC applications**
- **800V 30A DC Efuse design for Datacentre application**
- **JFET based Cascode topology design and development for SSCB**
- **Gate driver design for JFET Cascode topology**
- Experience in **Components selection and Schematic capture in ORCAD**
- Experience in **Proto BOM costing preparation and equivalent alternate component selection**
- Experience in **EMI/EMC Filter design & ESD, Surge and Lightning Protection design**
- Experience in **High Frequency Transformer and Magnetics design**
- Experience in **Gate driver design** for Low power and High-power application products
- **Micro controller and DSP Controller based battery charger design for Electric Vehicles (2W & 3W)**
- Experience in **Good knowledge in 3 phase BLDC Motor controller design**
- Experience in **AC & DC Current and Voltage sense circuit design**
- Experience in **LED Driver design for commercial and automotive products**
- Good knowledge in **Analog and Digital design**
- Good knowledge in **ADC selection, SPI and I2C** Communications
- Experience in **Solar Micro Inverter design**
- Experience in **Heat sink selection, thermal management and Enclosure design**
- Experience in **board power supply design for Set up box products**
- Experience in **Power budget preparation SoC Chipsets**
- Experience in **Power supply rails design for SoC Chipsets & WIFI Module**
- Experience in **Board power supply design for RF Front End Module (FEM), Memory Chipsets**
- Experience in **Battery charger design for 1 to 4 cells for 3S4P & 4S4P Battery packs**
- Experience in in **Battery management Systems (BMS) for 1 to 16S Battery pack**
- Experience in **UPS Design for Wifi -routers**
- Experience in **PoE & PoE ++ Converter design**

- In-depth knowledge in **RFQ proposal building like Effort Estimation, budgeting, resource allocation and project schedule**
- Experience to preparation of **RFP and RFI Proposals**
- Experience in **Product verification and validation (V&V) testing**
- Power Electronics converter simulation with **MATLAB, CASPOC, PSPICE, LT spice** tools
- **In-depth Knowledge in developing the products which compliance with various industry product standards and regulations like Industrial, Medical, Aerospace and Railways etc**
- Good experience to handling project of **DO-254 and DO-160** Standards
- Working knowledge in international standards like **UL489i,UL1449,IEC 60601-1, CISPR22, EN50155, EN50121, IEC60373, UL508**, **IEC 61508, ISO 26262**
- Experience in **PCB design Guideline setting, board file and Gerber file review**
- Experience in **board bring-up, functional & System level test plan preparation and testing**.
- Product **Functional Test Procedure and Test case preparation**
- Good knowledge in **FMEA, DFMEA**
- Good knowledge in **Product life cycle management**
- **BOM Obsolesce Management using by Silicon Expert tool**
- BOM Components **voltage, current and power stress calculation**
- **Pre-compliance & Compliance testing for EMI/EMC, Environmental, Shock & Vibration, Certification as per the product standards.**
- Expertise in using test equipment like **DSO, Power analyser, Spectrum analyser, DMM, Electronics Loads, High Voltage test equipment and Audio Analyser**
- Benchmarking and Value Engineering
- **Project effort estimation and schedule preparation**
- **Coordinating with different teams (HW, FW, PCB, Production and Mech) for smooth project execution**
- **Mentoring fresher and juniors in the team**
- Identifying and defining roles and responsibilities for project members
- Communicating project progress and hurdles to respective stakeholders and customer
- **Interacting with EMS vendor for PCB fabrication, Component procurement and prototyping**

Work Experience

#	**Designation**	**Service Period**	**Name of the Organization**
1	Sr, Design Engineer-Power Electronics	June 2025 -Present	Zepco Technologies Pvt Limited /ATOM Power Inc, Bangalore, Karnataka, India
2	Principal Engineer – Power Electronics HW	Feb.2024 – June 2025	Resonate Systems Pvt, Bangalore, Karnataka, India
3	Staff Engineer – Power Electronics HW	Apr.2022- Dec.2023	Dish Network Technologies, Bangalore, Karnataka, India
4	Asst.Project - Manager	Jan.2016- Apr.2022	Cyient Limited, Bangalore, Karnataka, India
5	Sr. Design Engineer - HW	Feb.2012 -Dec.2015	Sienna Ecad Technologies Pvt ltd, Bangalore, Karnataka, India
6	Design Engineer - HW	Oct.2010 - Jan.2012	Cooper Crouse -Hinds MTL India Pvt ltd, Chennai, Tamil Nadu, India
7	Design Engineer - HW	Sep.2008- Sep.2010	Wallaby Metering system Pvt ltd, Chennai, Tamil Nadu, India
8	Design Engineer -Projects	Nov.2007-Sep.2008	Vijay Electricals, Bangalore, Karnataka, India
9	Design Engineer (R&D)	July2006 - Nov.2007	S.M. Creative Electronics Pvt Ltd, Gurgaon, Haryana, India

Educational Qualification

#	Name of the Degree	Period	Name of the College & University
1	M.E - Power Electronics & Drives	2004-2006	Government College of Engineering, Tirunelveli, Affiliated to Anna University, Chennai, Tamil Nadu, India
2	B.E - Electrical and Electronics	2000-2003	Priyadarshini Engineering College, Vaniyambadi, Affiliated to Madras University, Chennai, Tamil Nadu, India
3	Diploma - Electrical and Electronics	1996-1999	Priyadarshini Polytechnic, Vaniyambadi, Affiliated to State Technical board Education, Tamil Nadu, India

Special Training

#	Name of the Training	Name of Intuition
1	CASPOC - Modelling and Simulation of Power Electronics converters	Govt. College of Engineering, Tirunelveli, Tamil Nadu, India
2	EMI/EMC Filter Design	Central Electronics Test Laboratory, Noida, Uttar Pradesh, India
3	Master Class on Solar PV System Design	Skill Dzire, Hyderabad, India
4	Master Class on EV Technology	Skill Dzire, Hyderabad, India
5	Project Management Skills for Leaders	LinkedIn Learnings

Project executed

1. 50W, Cab Audio Alarm unit

Description	Value
Input Voltage	16.8Vdc-137.8Vdc
Output Power & Output voltage	50W, 24V@2.1A
Efficiency	86%
Product compliance	EN 50155:2007, EN 50155:2017, IEC 60373:2007, EN50121
Protections	Under voltage, Overvoltage, over current, Thermal protection, Short circuit
Topology used	Buck-Boost, Isolated Fly back
PWM IC	UC3842, Switching frequency: 220KHz
Transformer core	EE25, Ferrite core,3F35, Ferrocube
Controller	ATSAMV71Q20B-AAB, ARM Coretex-M7
Audio amplifier	TAS5720M
No. of logic input	16
Transformer core	EE25, Ferrite core,3C90, Ferro cube
Software	ORCAD, Multisim, LTspice,
Communication	RS485, RS232, USB
Application	Railways
Role	Architect and Hardware Lead
Team size	4
Project duration	1 Year
Name of the Organization \ Client	Cyient Ltd, Bangalore

2. 1.5KW BLDC Motor Driver Controller

Description	Value
Input Voltage	54-V typ (9V min to 63-V max)
Output Power & Output voltage	1500W
BLDC No. of Phases	3
RMS Winding Current	28A
Peak Winding Current	400A
Control method	Unipolar trapezoidal PWM
Inverter Switching frequency	20KHZ
Efficiency	99%
Feedback Signals	DC bus voltage, Motor phase voltages, Motor position Hall sensors, Low side DC bus current
Protections	Cycle-by-cycle over current, input under-voltage, over temperature, MOSFET gate-source open circuit and short circuit
Cooling	Natural Cooling
Driver IC	DRV8320
Speed Control	0-100K RPM
Controller	MSP430FR2355
Software	ORCAD, LTspice,
Communication	UART
Application	Vacuum Cleaner
Operating Temperature	‘-40 to 55 C
Role	Architect and Hardware Lead
Team size	3
Project duration	9 Months
Name of the Organization \ Client	Cyient Ltd, Bangalore

3. 3.5 KW Battery Charger

Description	Value
Input Voltage	90Vac-265Vac ,50Hz, Single phase
Output Power & Output voltage	3.5KW, 89V/28A, 13.5V/74A
Input PF & THD	0.99(max) & 5%
Efficiency	84%
PFC Converter Topology used	Interleaved Boost converter
DC-DC Converter Topology used	Phase shifted full bridge
PFC PWM IC & Switching frequency	UCC28070 & 100KHz
PFC PWM IC & Switching frequency	UCC28950 & 60KHz
DSP Controller	dsPIC33FJ16GS506
Product compliance	CISPR22 Class B
Protections	Overvoltage, over current, Thermal protection, short circuit
Transformer core	ETD59, Ferritecore,3c90, Ferroxcube
Software	ORCAD, LTspice, Pspice
Application	Electric Car battery charger
Role	Architect and Hardware Lead
Team size & Project duration	4 & 1.5 Year
Name of the Organization \ Client	Cyient Ltd, Bangalore

4. 374W Battery charger

Description	Value
Input Voltage	90Vac-267Vac,1phase,50Hz
Output Power	374W
Output Boost Voltage	17Vdc@22A
Output Float voltage	13.8Vdc@2.5A
Input power factor	0.99(max)
Efficiency	90%
Product compliance	CISPR22ClassB
Protections	Overvoltage, over current, Thermal protection
PFC IC	PFS7329H
PFC Switching Frequency	66KHz
DC-DC Converter side Topology used	LLC Converter
Switching frequency	190-300KHz
LLCIC	LCS708HG
Transformer core	ETD39, Ferritecore,3c90, ferroxcube
Software	ORCAD, PI Software
Application	Battery charger for 150Ah Battery
Role	Architect and Hardware Lead
Team size & Project duration	2 & 9 months
Name of the Organization \ Client	Sienna Ecad, Bangalore \ Exide Industries

5. 800W UPS (Off-line)

Description	Value
Input Voltage	10.4Vdc-17Vdc
Output Power & Output voltage	800W, 180-230Vac,4.44A,49-51Hz
Output PF & THD	0.8(max) & 5%
Output waveform	Pure Sine wave
Output power factor	0.8(max)
Efficiency	84%
Product compliance	CISPR22 Class B
Protections	Overvoltage, over current, Thermal protection
DC-DC Converter side Topology used	Push-Pull
PWM IC & Switching frequency	SG3525A & 20KHz
Inverter side Topology used	Full Bridge
Switching Frequency	20KHz
DSP Controller	dsPIC33FJ16GS504
Transformer core	ETD54, Ferritecore,3c90, Ferroxcube
Software	ORCAD, Multisim, Spice
Application	Home Applications
Role	Architect and Hardware Lead
Team size & Project duration	2 & 1 Year
Name of the Organization \ Client	Sienna Ecad, Bangalore \ Exide Industries

6.220W Solar Micro Inverter (Off-Grid)

Description	Value
DC-DC Converter side Topology	Interleaved Fly back
Input Voltage	24Vdc–55Vdc
Output Power	220W, 230Vac,0.95A,49-51Hz
Output PF & THD	0.8(max) & 5%
Output waveform	Pure Sine wave
Efficiency	90%
Product compliance	CISPR22ClassB
Protections	Over voltage, over current, Thermal protection

Switching frequency	40KHz
Inverter side Topology used	Full Bridge
Switching Frequency	40KHz
DSP Controller	dsPIC33FJ16GS504
Transformer core	ETD34, Ferritecore,3c90, Ferroxcube
Software	ORCAD, Multisim, Spice
Application	Home Applications
Role	Architect and Hardware Lead
Team size & Project duration	2 & 6 Months
Name of the Organization \ Client	Sienna Ecad, Bangalore \ On Semi

7.230VAC Power Line Surge Protective Devices (SPD)

Description	**Value**
Input Voltage	230vac ,50Hz
Max leakage current	5μA
Maximum surge current	20kA (8/20μs waveform) per line
Maximum rated load current	10A
Bandwidth	25 kHz
Response time	1ns
Series resistance	< 1Ω (Very low)
Working temperature	-40 to 80 C
Role	Project lead
Team size & Project duration	1 & 9 months
Name of the Organization \ Client	Cooper Crouse – Hinds MTL India Pvt ltd, Chennai.

8.10W, Multi-output, AC-DC Power supply

Description	**Value**
Input Voltage	90Vac-267Vac,1phase,50Hz
Output Power & Output voltage	10W, 3V3@0.1A, 5V@0.2A, 12V@0.5A
Input power factor	0.8(max)
Efficiency	80%
Product compliance	CISPR22 Class B
Protections	Over voltage, over current, Thermal protection
Topology used	Fly back
PWM IC & Switching frequency	UC3843 & 66KHz
Transformer core	E16, Ferrite core, N87, EPCOS
Software Tools used	ORCAD, Caspoc
Application	Energy meter auxiliary power supply
Role	Project lead
Team size & Project duration	1 & 2 Months
Name of the Organization \ Client	Wallaby Metering Systems, Chennai.

9.240W, AC-DC Power supply

Description	**Value**
Input Voltage	90Vac-267Vac,1phase,50Hz
Output Power & Output voltage	240W, 48V@5A
Input power factor	0.99(max)
Efficiency	88%
Product compliance	CISPR22 Class B
Protections	Over Voltage, over Current, Thermal protection
Topology used	Forward
PWM IC & Switching frequency	UC3843 & 66KHz
Power factor control IC	NCP1650(On semiconductor)
Transformer core	ETD34, Ferritecore,3c90, Ferroxcube

Software Tools used	ORCAD, Caspoc
Application	Telecom power supply
Role	Project lead
Team size & Project duration	2 & 6 months
Name of the Organization \ Client	Wallaby metering systems, Chennai.

10.150W AC-DC Power supply

Description	Value
Input Voltage	90Vac-267Vac,1phase,50Hz
Output Power & Output voltage	150W, 24@6.25A
Input power factor	0.8(max)
Efficiency	85%
Product compliance	CISPR22ClassB
Protections	Over voltage, over current, Thermal protection
Topology used	Fly back
PWM IC & Switching frequency	UC3845 & 75KHz
Transformer core	ETD29, Ferritecore,3c90, Ferroxcube
Software Tools used	ORCAD, Caspoc
Application	Panel board power supply
Role	Project lead
Team size & Project duration	2 & 6 months
Name of the Organization \ Client	Wallaby metering systems, Chennai.

11.204W, Multi-output, AC-DC Power supply

Description	Value
Input Voltage	90Vac-267Vac,1phase,50Hz
Output Power & Output voltage	204W, 24@6.5A,12V@4A
Input power factor	0.99(max)
Efficiency	88%
Product compliance	CISPR22 Class B
Protections	Over voltage, over current, Thermal protection
Topology used	Forward
PWM IC & Switching frequency	UC3843 & 66KHz
Power factor control IC	NCP1650(On semiconductor)
Transformer core	ETD34, Ferritecore,3c90, Ferroxcube
Software	ORCAD, Caspoc
Application	Panel board power supply
Role	Project lead
Team size & Project duration	2 & 6 months
Name of the Organization \ Client	Wallaby metering systems, Chennai.

12.100W AC-DC Power supply

Description	Value
Input Voltage	90Vac-267Vac,1phase,50Hz
Output Power & & Output voltage	100W, 3.3@20A,5V@2A,12V@2A
Input power factor	0.99(max)
Efficiency	87%
Product compliance	CISPR 22Class B
Protections	Over voltage, over current, Thermal protection
Topology used	Fly back
PWMIC & Switching frequency	UC3845 & 75KHz
Power factor control IC	NCP1650(On semiconductor)

Transformer core	ETD34, Ferritecore,3c90, Ferroxcube
Software	ORCAD
Application	Network power supply
Role	Project lead
Team size & Project duration	2 & 6 months
Name of the Organization \ Client	S.M. Creative Electronics Pvt Ltd, Gurgaon

13.1092W, Multi-output, AC-DC Power supply

Description	Value
Input Voltage	180Vac-267Vac,1phase,50Hz
Output Power & Output voltage	1092W, 3V3@40A,12V@80A
Input power factor	0.8(max)
Efficiency	88%
Product compliance	MIL 461E
Protections	Overvoltage, over current, Thermal protection
Topology used	Full Bridge
PWM IC	SG3525A, Switching frequency: 66KHz
Vicor IC	V48a3V3c264Bl for3.3V output, V48a12c500Bl for12V
Transformer core	ETD59, Ferrite core,3C90, Ferrocube
Software Tools used	ORCAD, Caspoc
Transformer core	ETD59, Ferrite core,3C90, Ferrocube
Application	Military Network power supply
Role	Project lead
Team size & Project duration	2 & 10 Months
Name of the Organization \ Client	S.M. Creative Electronics Pvt Ltd, Gurgaon

PERSONAL DETAILS

Name K.B. SENDHIL KUMAR

Phone +91 8123867417

E-Mail **sendhilbalu@gmail.com**

sendhil_balu@yahoo.co.in

Date of Birth 2nd June 1978

Nationality Indian

Marital status Married

Postal Address 25 & 26, S004, Second floor,1st main,

Pattalamma Layout, Kadugodi

Bangalore-67

Karnataka, India.

DECLARATION:

I declare that the details furnished above are true and correct to the best of my knowledge.

Date: Yours Sincerely,

Place: Bangalore **K.B. SENDHIL KUMAR**